ALGEBRA 2

Extra Practice — Chapter 1 • Practice Set 1

Name: _____ Date: _____ Period: _____

SHOW ALL WORK.

1. Solve $\frac{3}{4}(3x-4)-\frac{1}{6}(2x-5)=\frac{1}{2}(4x+1)$. *(4 points)*

2. Find the value of r so that the line passes through $\left(\frac{1}{4},-\frac{1}{2}\right)$ and $\left(r,\frac{3}{4}\right)$ with the slope $\frac{5}{8}$. *(4 points)*

$r =$: _____

3 – 4. Write an equation of the line that satisfies the given conditions and identify its rate of change.

3. Passing through $(-3,5)$ and $(1,-3)$ *(5 points)*

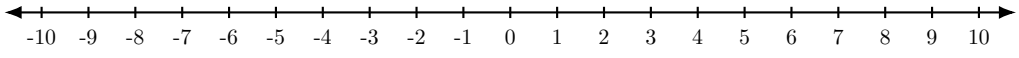
Rate of change: _____

4. Passing through $(2,-1)$ and perpendicular to $2x+4y=8$ *(5 points)*

Rate of change: _____

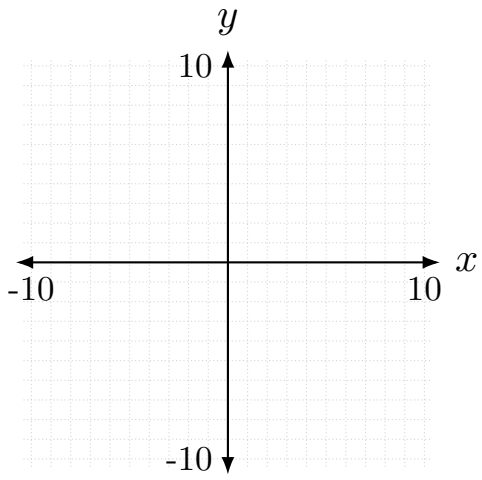
5. Solve $5-3|2x-1|>-4$, graph the inequality, and write the inequality in interval notation. *(6 points)*

Inequality: _____



Interval notation: _____

6. Graph $y - 3 = \frac{1}{2}(x - 4)$. Then find the slope and the y-intercept. *(6 points)* **Slope:** _____ **y-intercept:** _____



ALGEBRA 2

Extra Practice — Chapter 1 • Practice Set 2

Name: _____ Date: _____ Period: _____

SHOW ALL WORK.

1. Solve $\frac{3}{4}(3x-5)-\frac{1}{2}(x-3)=\frac{1}{2}(4x-2)$. *(4 points)*

2. Find the value of r so that the line passes through $\left(\frac{1}{3},-\frac{1}{6}\right)$ and $\left(r,\frac{5}{6}\right)$ with the slope $\frac{2}{3}$. *(4 points)*

r **=:** _____

3 – 4. Write an equation of the line that satisfies the given conditions and identify its rate of change.

3. Passing through $(0,-3)$ and $(4,5)$ *(5 points)*

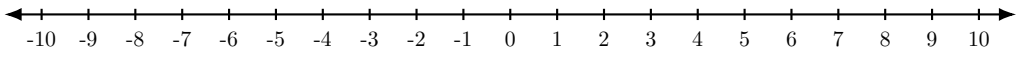
Rate of change: _____

4. Passing through $(-4,1)$ and perpendicular to $3x-6y=12$ *(5 points)*

Rate of change: _____

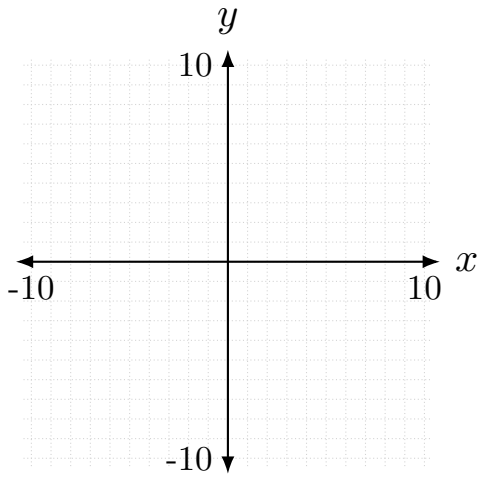
5. Solve $7-2|3x+1|\geq -3$, graph the inequality, and write the inequality in interval notation. *(6 points)*

Inequality: _____



Interval notation: _____

6. Graph $y + 2 = -3(x - 1)$. Then find the slope and the y-intercept. *(6 points)* **Slope:** _____ **y-intercept:** _____



ALGEBRA 2

Extra Practice — Chapter 1 • Practice Set 3

Name: _____ Date: _____ Period: _____

SHOW ALL WORK.

1. Solve $\frac{5}{6}(3x-2)-\frac{1}{2}(2x-1)=\frac{1}{3}(6x+4)$. *(4 points)*

2. Find the value of r so that the line passes through $\left(-\frac{1}{2},\frac{1}{4}\right)$ and $\left(r,-\frac{3}{4}\right)$ with the slope $-\frac{5}{6}$. *(4 points)*

   $r =$**:** _____

3 – 4. Write an equation of the line that satisfies the given conditions and identify its rate of change.

3. Passing through $(4,-1)$ and $(-4,3)$ *(5 points)*

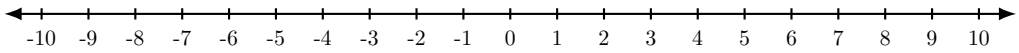
   Rate of change: _____

4. Passing through $(2,3)$ and perpendicular to $5x+2y=10$ *(5 points)*

   Rate of change: _____

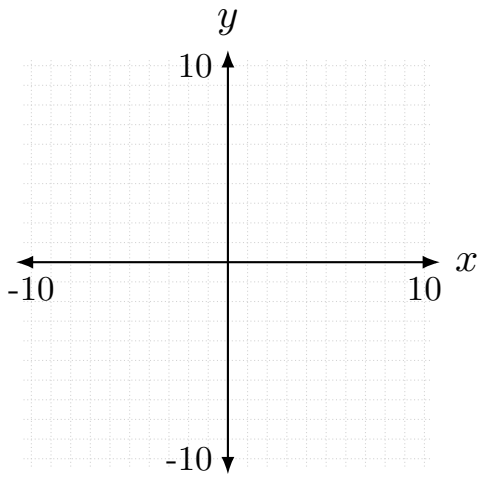
5. Solve $6-4|2x-3|>-10$, graph the inequality, and write the inequality in interval notation. *(6 points)*

   Inequality: _____



Interval notation: _____

6. Graph $y - 1 = -\frac{2}{3}(x + 3)$. Then find the slope and the y-intercept. *(6 points)* **Slope:** _____ **y-intercept:** _____



ANSWER KEY

Algebra 2 — Extra Practice, Chapter 1

Practice Set 1 — Answer Key

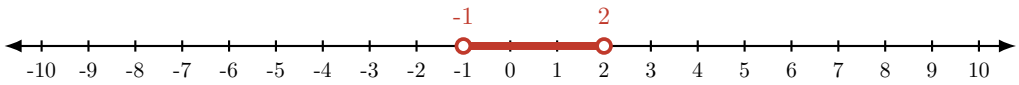
1. Distribute: $2x - \frac{8}{3} - \left(\frac{1}{3}x - \frac{5}{6}\right) = 2x + \frac{1}{2} \;\rightarrow$ Combine: $\frac{5}{3}x - \frac{11}{6} = 2x + \frac{1}{2} \;\rightarrow\; -\frac{1}{3}x = \frac{1}{2} + \frac{11}{6} = \frac{7}{3}$
   Answer: $x = -7$

2. $\dfrac{5}{8} = \dfrac{\frac{3}{4} - (-\frac{1}{2})}{r - \frac{1}{4}} = \dfrac{5/4}{r - 1/4} \;\rightarrow\; r - \frac{1}{4} = \frac{5}{4} \div \frac{5}{8} = 2$
   Answer: $r = \dfrac{9}{4}$

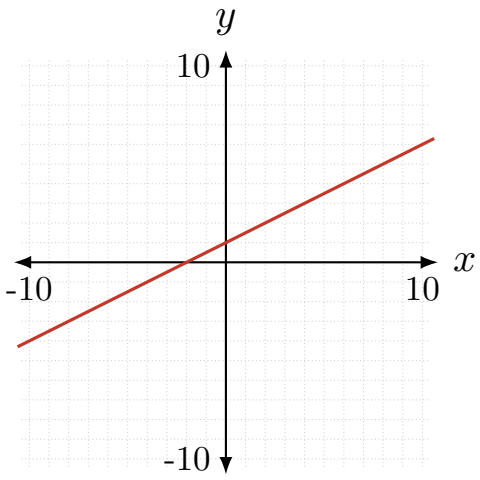
3. $m = \dfrac{-3-5}{1-(-3)} = \dfrac{-8}{4} = -2 \;\rightarrow\; y - 5 = -2(x+3) \;\rightarrow\; y = -2x - 1$
   Answer: $y = -2x - 1$; **rate of change** $= -2$

4. $2x + 4y = 8 \rightarrow y = -\frac{1}{2}x + 2$, given slope $= -\frac{1}{2} \;\rightarrow$ perpendicular slope $= 2$; $y - (-1) = 2(x - 2)$
   Answer: $y = 2x - 5$; **rate of change** $= 2$

5. $-3|2x - 1| > -9 \rightarrow |2x - 1| < 3 \rightarrow -3 < 2x - 1 < 3 \rightarrow -2 < 2x < 4$
   Answer: $-1 < x < 2$; **interval notation** $(-1, 2)$



6. $y = \frac{1}{2}x - 2 + 3 = \frac{1}{2}x + 1$
   Answer: **slope** $= \dfrac{1}{2}$; y**-intercept** $= 1$



Practice Set 2 — Answer Key

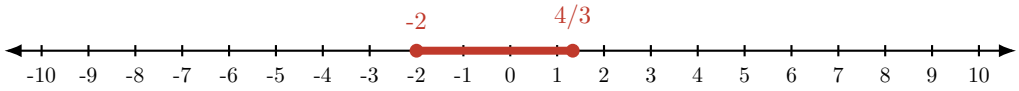
1. Distribute: $\frac{9}{4}x - \frac{15}{4} - \left(\frac{1}{2}x - \frac{3}{2}\right) = 2x - 1 \;\rightarrow$ Combine: $\frac{7}{4}x - \frac{9}{4} = 2x - 1 \;\rightarrow\; -\frac{1}{4}x = -1 + \frac{9}{4} = \frac{5}{4}$
   Answer: $x = -5$

2. $\dfrac{2}{3} = \dfrac{\frac{5}{6} - (-\frac{1}{6})}{r - \frac{1}{3}} = \dfrac{1}{r - 1/3} \;\rightarrow\; r - \frac{1}{3} = \frac{3}{2}$
   Answer: $r = \dfrac{11}{6}$

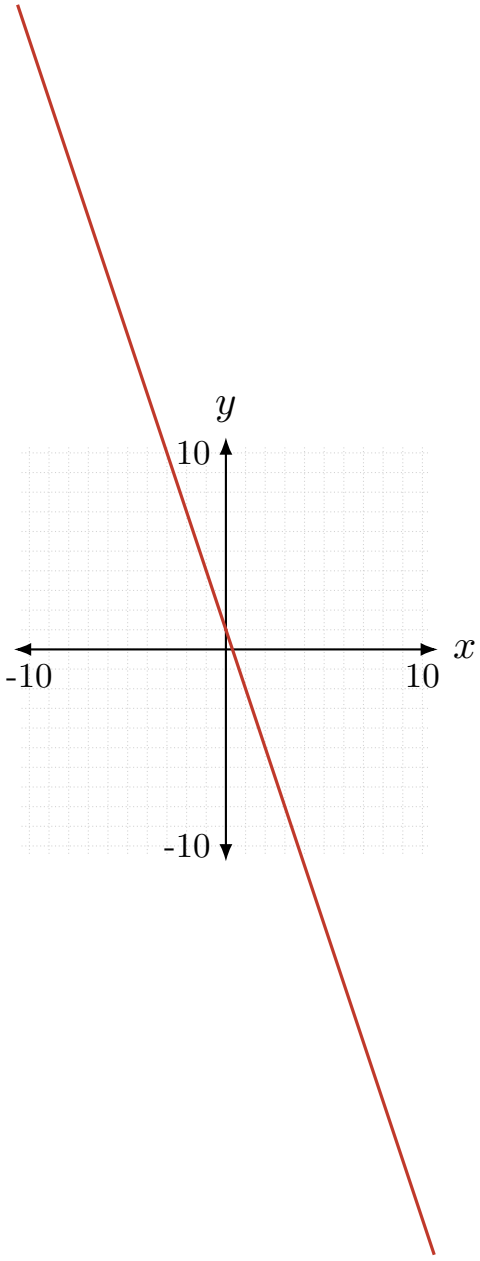
3. $m = \dfrac{5 - (-3)}{4 - 0} = \dfrac{8}{4} = 2 \;\rightarrow\; y - (-3) = 2(x - 0) \;\rightarrow\; y = 2x - 3$
   Answer: $y = 2x - 3$; **rate of change** $= 2$

4. $3x - 6y = 12 \rightarrow y = \frac{1}{2}x - 2$, given slope $= \frac{1}{2} \;\rightarrow$ perpendicular slope $= -2$; $y - 1 = -2(x + 4)$
   Answer: $y = -2x - 7$; **rate of change** $= -2$

5. $-2|3x + 1| \geq -10 \rightarrow |3x + 1| \leq 5 \rightarrow -5 \leq 3x + 1 \leq 5 \rightarrow -6 \leq 3x \leq 4$
   Answer: $-2 \leq x \leq \dfrac{4}{3}$; **interval notation** $\left[-2, \dfrac{4}{3}\right]$



6. $y = -3x + 3 - 2 = -3x + 1$
   Answer: **slope** $= -3$; y**-intercept** $= 1$


y
10
-10
10
x
-10

Practice Set 3 — Answer Key

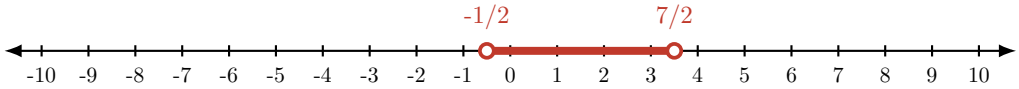
1. Distribute: $\frac{5}{2}x - \frac{5}{3} - \left(x - \frac{1}{2}\right) = 2x + \frac{4}{3}$ $\rightarrow$ Combine: $\frac{3}{2}x - \frac{7}{6} = 2x + \frac{4}{3}$ $\rightarrow$ $-\frac{1}{2}x = \frac{4}{3} + \frac{7}{6} = \frac{5}{2}$
   Answer: $x = -5$

2. $-\dfrac{5}{6} = \dfrac{-\frac{3}{4} - \frac{1}{4}}{r - \left(-\frac{1}{2}\right)} = \dfrac{-1}{r + 1/2}$ $\rightarrow$ $r + \frac{1}{2} = -1 \div \left(-\frac{5}{6}\right) = \frac{6}{5}$
   Answer: $r = \dfrac{7}{10}$

3. $m = \dfrac{3 - (-1)}{-4 - 4} = \dfrac{4}{-8} = -\dfrac{1}{2}$ $\rightarrow$ $y - (-1) = -\frac{1}{2}(x - 4)$ $\rightarrow$ $y = -\frac{1}{2}x + 1$
   Answer: $y = -\dfrac{1}{2}x + 1$; **rate of change** $= -\dfrac{1}{2}$

4. $5x + 2y = 10 \rightarrow y = -\frac{5}{2}x + 5$, given slope $= -\frac{5}{2}$ $\rightarrow$ perpendicular slope $= \frac{2}{5}$; $y - 3 = \frac{2}{5}(x - 2)$
   Answer: $y = \dfrac{2}{5}x + \dfrac{11}{5}$; **rate of change** $= \dfrac{2}{5}$

5. $-4|2x - 3| > -16 \rightarrow |2x - 3| < 4 \rightarrow -4 < 2x - 3 < 4 \rightarrow -1 < 2x < 7$
   Answer: $-\dfrac{1}{2} < x < \dfrac{7}{2}$; **interval notation** $\left(-\dfrac{1}{2}, \dfrac{7}{2}\right)$



6. $y = -\frac{2}{3}x - 2 + 1 = -\frac{2}{3}x - 1$
   Answer: **slope** $= -\dfrac{2}{3}$; **y-intercept** $= -1$

